

UBT408: GENETIC & METABOLIC ENGINEERING

L	T	P	Cr
3	0	0	3.0

Course Objective: This course would familiarize Students with facile molecular techniques involved in isolation and manipulation of genetic material for achieving the desired goal.

Syllabus

Introduction: Scope of genetic engineering, Restriction enzymes and other DNA modifying enzymes used in cloning and other DNA manipulations, essential and desired properties of cloning vectors, examples of cloning and expression vectors, recombinant protein expression in bacteria, yeast and higher eukaryotic systems, rDNA technology in purification of overexpressed recombinant proteins

Genomic and cDNA libraries: Construction and screening of Genomic and cDNA libraries, expression and regulation studies of genes, DNA labelling, nucleic acid hybridization

Applications of genetic engineering: DNA and protein sequencing, Polymerase Chain Reactions (PCR), Site-directed mutagenesis and applications, Molecular markers, Detecting protein-protein interactions, High-throughput techniques, Gene therapy, Genome editing, DNA fingerprinting, RFLP and RAPD

Metabolic Engineering: Introduction, Molecular strategies for rerouting of metabolic pathways in microbes, plants and animals, Various case studies, directed production of novel molecules in microbes and other organisms having therapeutic and industrial values, selection of host-vector system, constraints of the process, genetic instability and host cell mutations

Course Learning Objectives (CLO)

The students will be able to:

1. Design and perform cloning of foreign DNA in an appropriate vector.
2. Select the suitable hosts for a given vector and apply it for cloning or expression of recombinant proteins.
3. Amplify DNA with a target sequence using polymerase chain reaction.
4. Demonstrate application of recombinant DNA technology in various research, industrial, diagnostic and therapeutic applications.
5. Design process for enhanced production of secondary metabolites.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Text Books

1. Primrose, S.B. and Twyman, R.M., Principles of Gene Manipulation and Genomics, Blackwell Publishing (2006).
2. J. E. Krebs, E. S. Goldstein, S. T. Kilpatrick, Lewin's Genes XI, International Edition, Pearson Education (2014).
3. Rastogi, S. & Pathak, N., Genetic Engineering, Oxford Higher Education (2009).

Reference Books

1. Alberts, B., Johnson, A., Lewis, J., Raff, M., Roberts, K. and Walter, P., Molecular Biology of the Cell, 5th Edition, Garland Science Publishing (2008).
2. Fritsch, J. and Maniatis, E.F., Molecular Cloning, A laboratory Manual, Cold Spring Harbor Laboratory (1999).

Evaluation scheme

S. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30

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